Performance Evaluation of AOMDV Protocol in Wired and Wireless Networks using ns-3

1 Introduction

This report presents a performance evaluation of the AOMDV (Ad hoc On-demand Multipath Distance Vector) routing protocol using the ns-3 network simulator. The study analyzes both wired and wireless network scenarios under varying parameters to understand the behavior of the protocol under different network conditions.

The key performance metrics evaluated include throughput, end-to-end delay, packet delivery ratio (PDR), packet drop ratio, and energy consumption.

2 Network Topologies Under Simulation

Two different network environments were considered:

2.1 Wireless Network

- Multi-hop ad hoc network
- Nodes deployed with mobility
- AOMDV routing protocol used

2.2 Wired Network

- Static topology
- Point-to-point links
- AOMDV applied for routing

3 Parameters Under Variation

The following parameters were varied:

- Number of nodes
- Number of flows
- Packets per second (PPS)
- Node speed (wireless only)

3.1 Note on Parameter Selection

The originally specified parameters were:

- Nodes: 20, 40, 60, 80, 100
- Flows: 10, 20, 30, 40, 50
- PPS: 100, 200, 300, 400, 500

4 Modifications Made in the Simulator

The following modifications were implemented:

- A new **AOMDV routing module** was implemented from scratch under `src/aomdv`, including helper, routing, packet, and routing-table components integrated with the ns-3 IPv4 routing framework.
- Custom control packet headers were implemented for **RREQ**, **RREP**, **RERR**, and **HELLO** messages, with serialization/deserialization support and explicit message typing.
- Multipath routing state was added by storing multiple next hops per destination (with sequence numbers, hop counts, and expiry times), selecting the best valid path for forwarding, and purging expired paths periodically.
- Route discovery/maintenance logic was implemented with duplicate RREQ suppression, reverse-path installation from RREQs, forward-path installation from RREPs, and route error propagation/invalidations when links fail.
- Per-interface UDP sockets and timers were used to exchange AOMDV control traffic, with configurable attributes such as maximum paths, active route timeout, hello interval, and path discovery time.

5 Results

5.1 Wireless Network Results

5.1.1 Effect of PPS

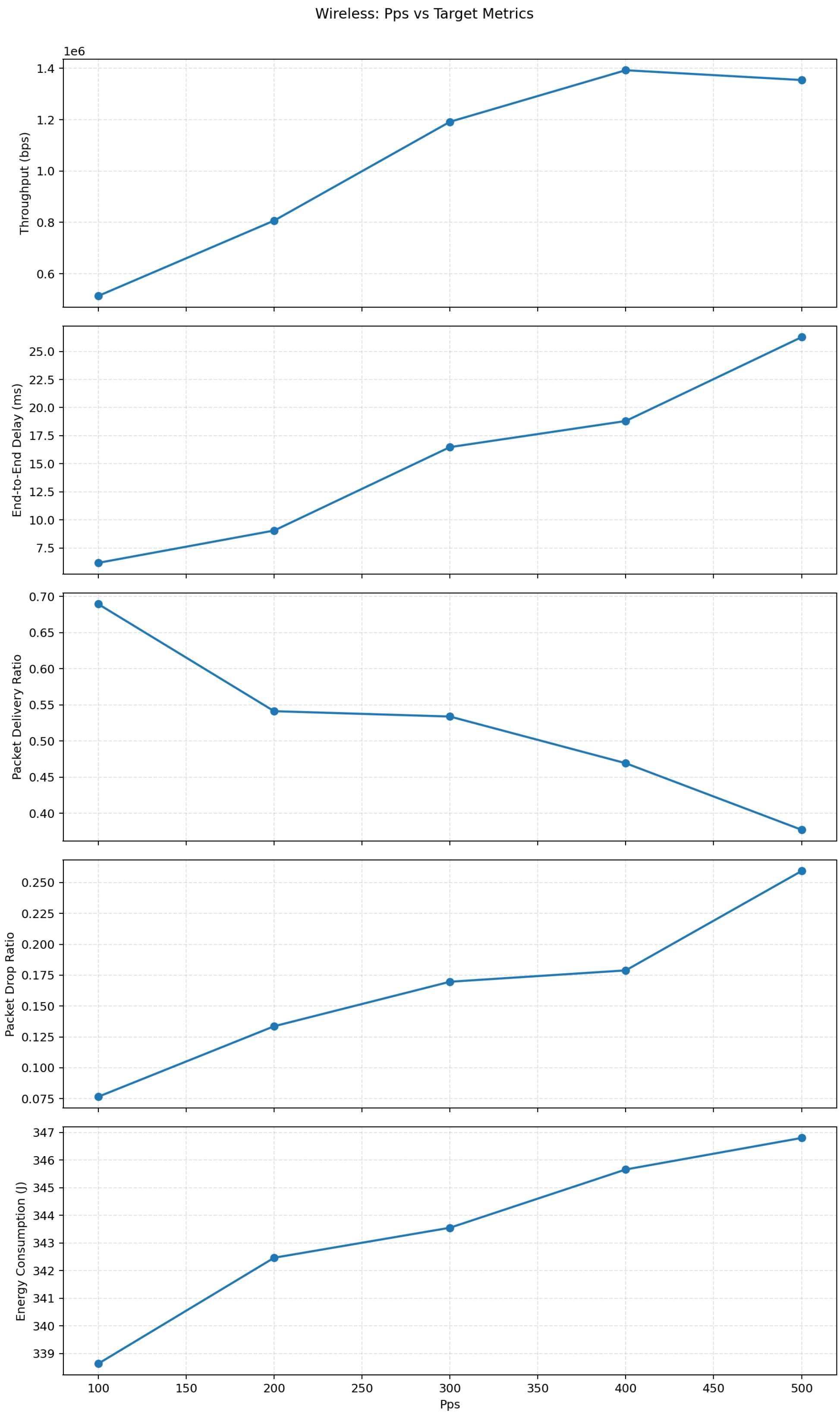

Figure 1: Wireless: Performance vs PPS

Throughput increases with PPS initially, reaching a peak before slightly declining due to network congestion. Delay and packet drop ratio increase steadily with higher PPS, while PDR decreases. Energy consumption also increases with load.

5.1.2 Effect of Number of Flows

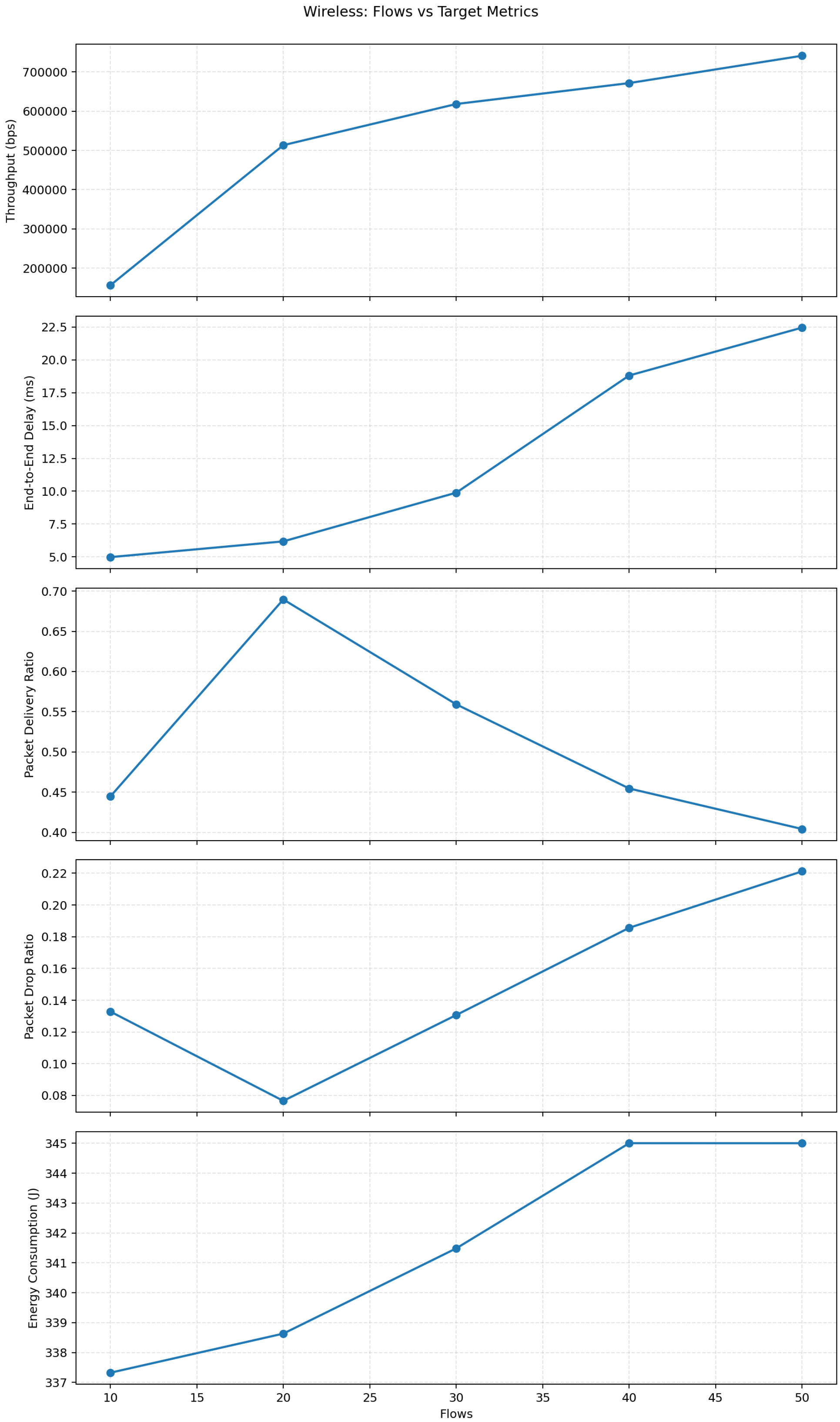

Figure 2: Wireless: Performance vs Number of Flows

Throughput peaks at moderate flow levels and decreases as contention increases. Delay rises significantly at higher flows, indicating congestion. Packet delivery ratio drops sharply, while packet drops increase.

5.1.3 Effect of Node Speed

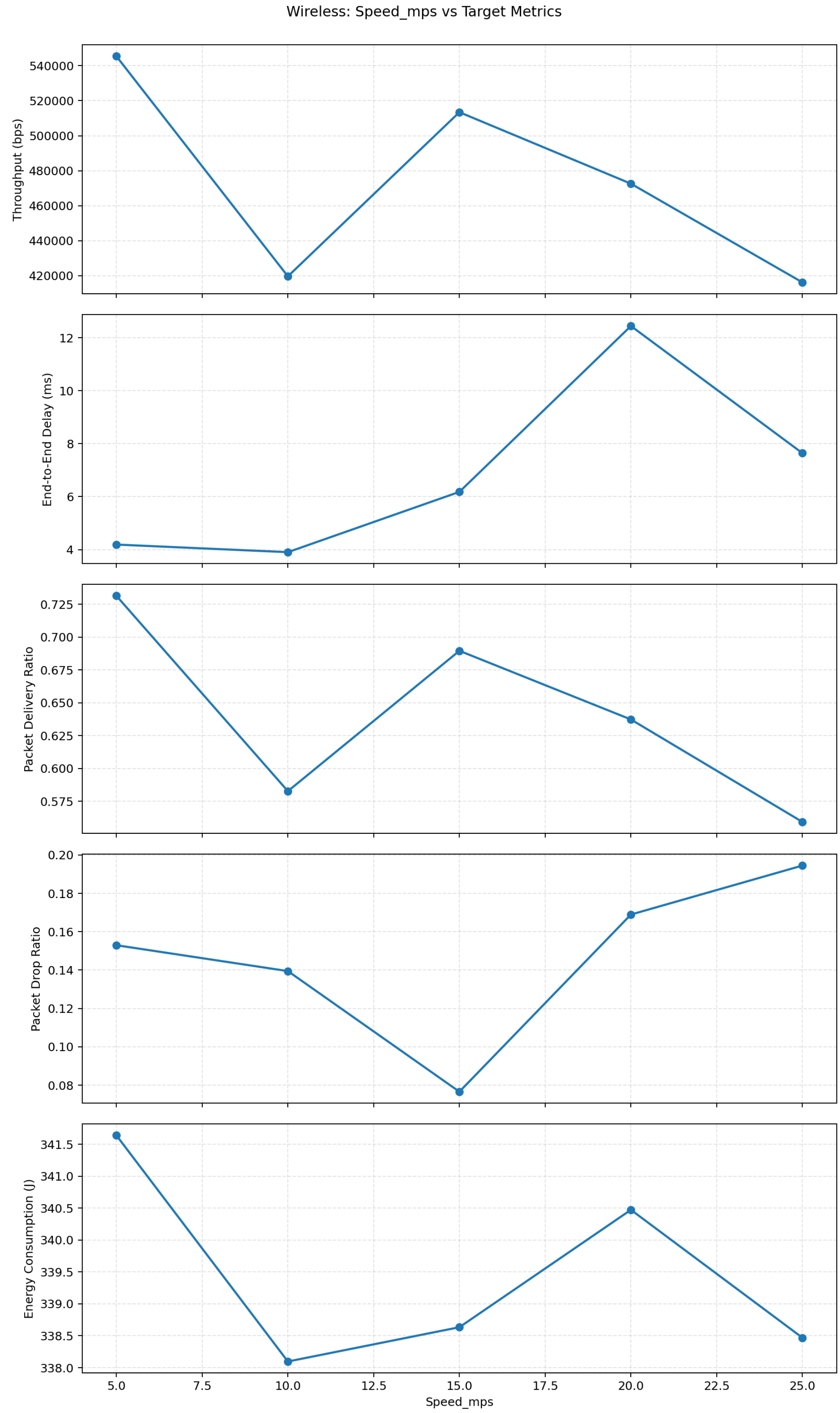

Figure 3: Wireless: Performance vs Node Speed

Higher mobility leads to unstable routes, reducing throughput and PDR. Delay and packet loss increase due to frequent route breakages.

5.1.4 Effect of Number of Nodes

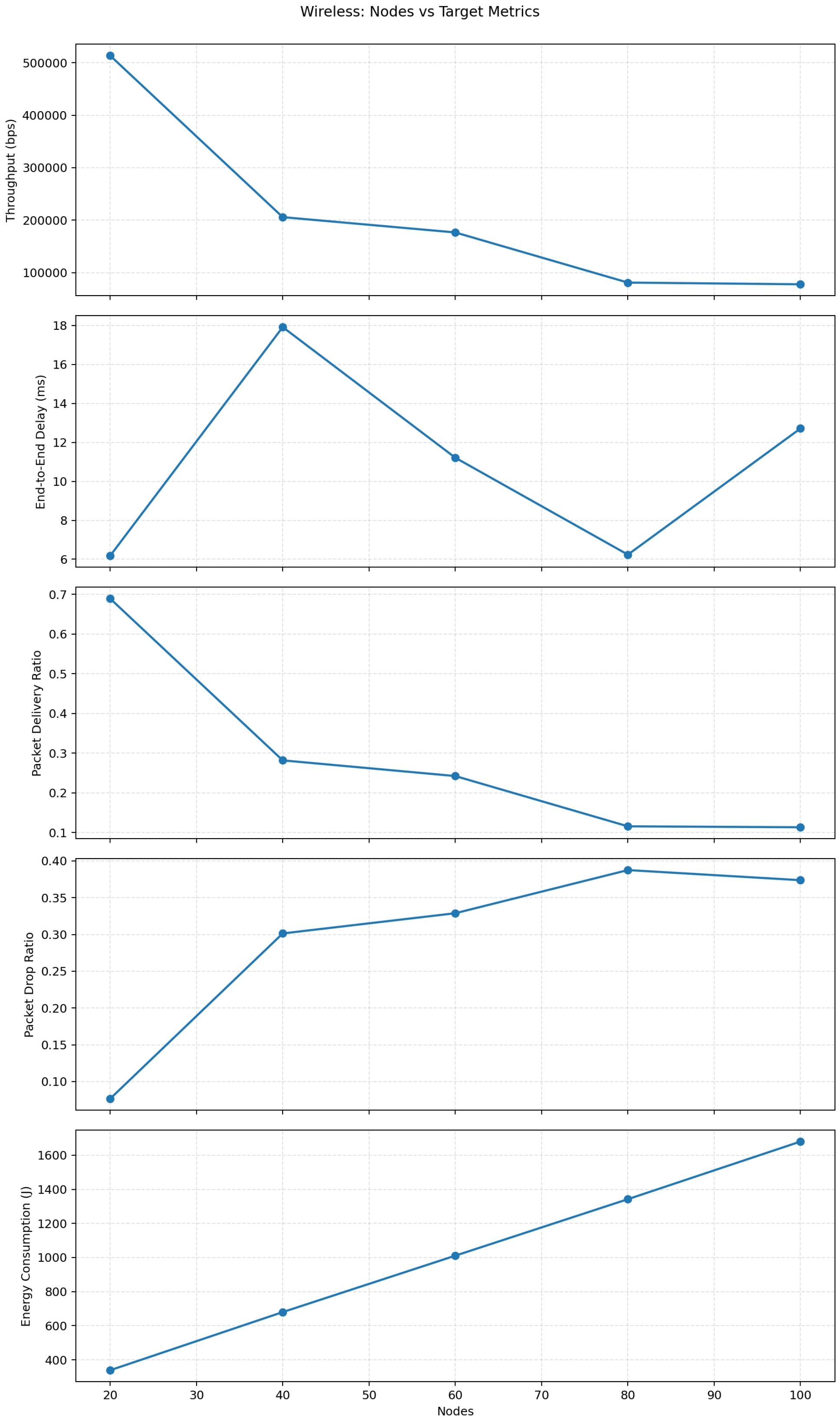

Figure 4: Wireless: Performance vs Number of Nodes

Increasing node density initially improves connectivity but eventually leads to congestion, reducing throughput and PDR while increasing delay and energy consumption.

5.2 Wired Network Results

5.2.1 Effect of PPS

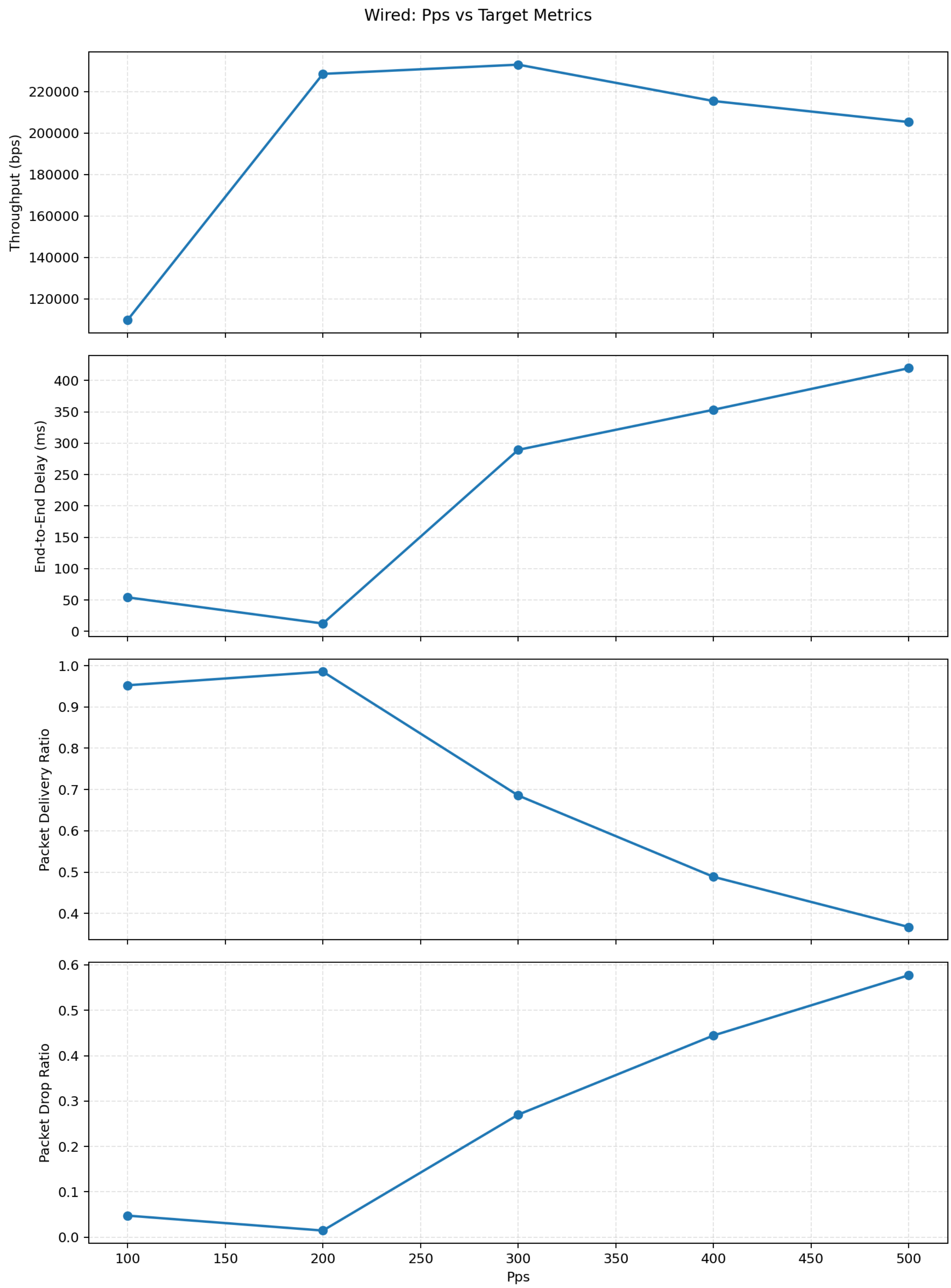

Figure 5: Wired: Performance vs PPS

Throughput increases up to a threshold and then drops sharply due to buffer overflow and congestion. Delay increases significantly at higher PPS values.

5.2.2 Effect of Number of Flows

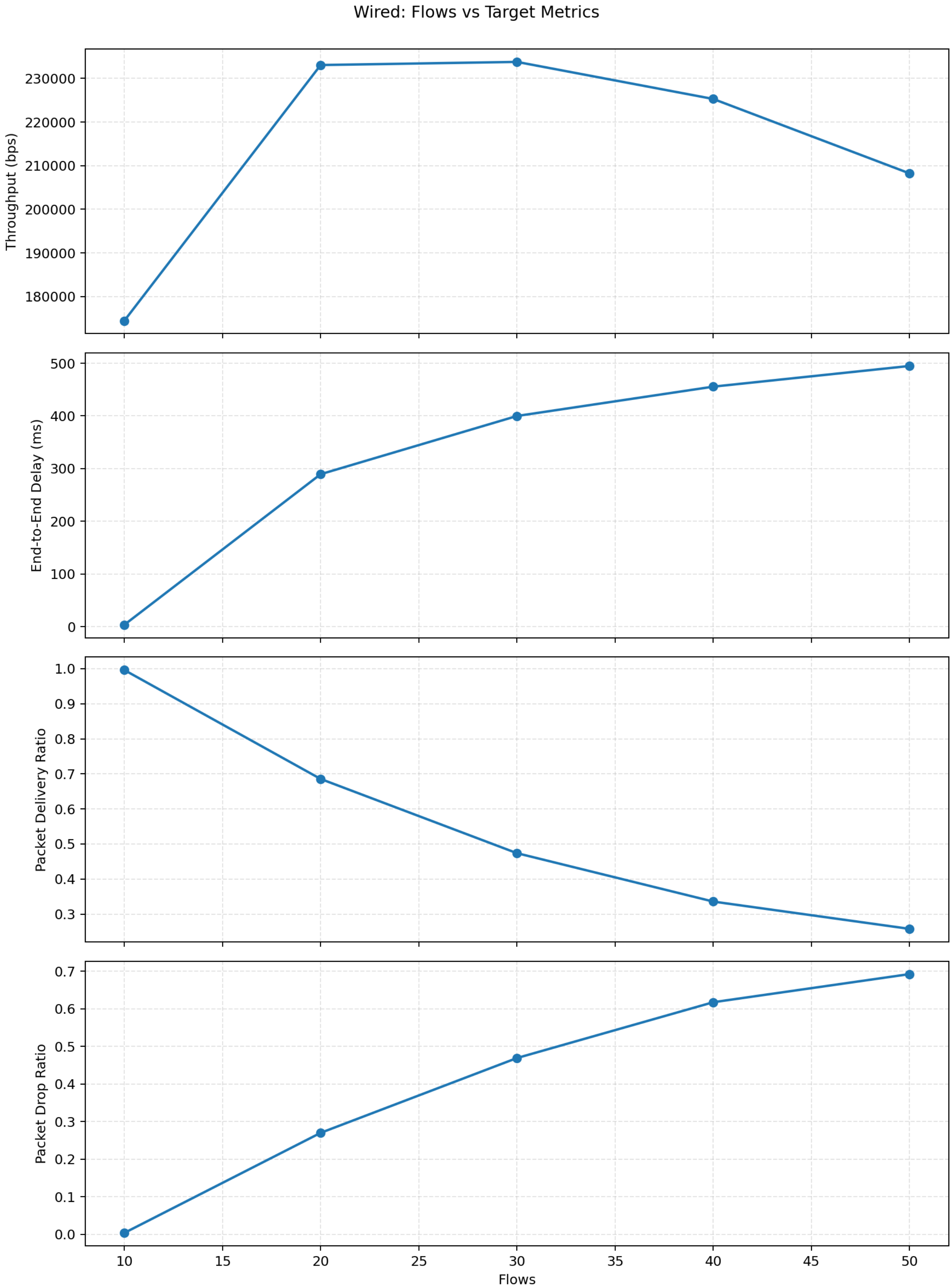

Figure 6: Wired: Performance vs Number of Flows

As flows increase, throughput decreases due to contention. Delay and packet drop ratio increase significantly.

5.2.3 Effect of Number of Nodes

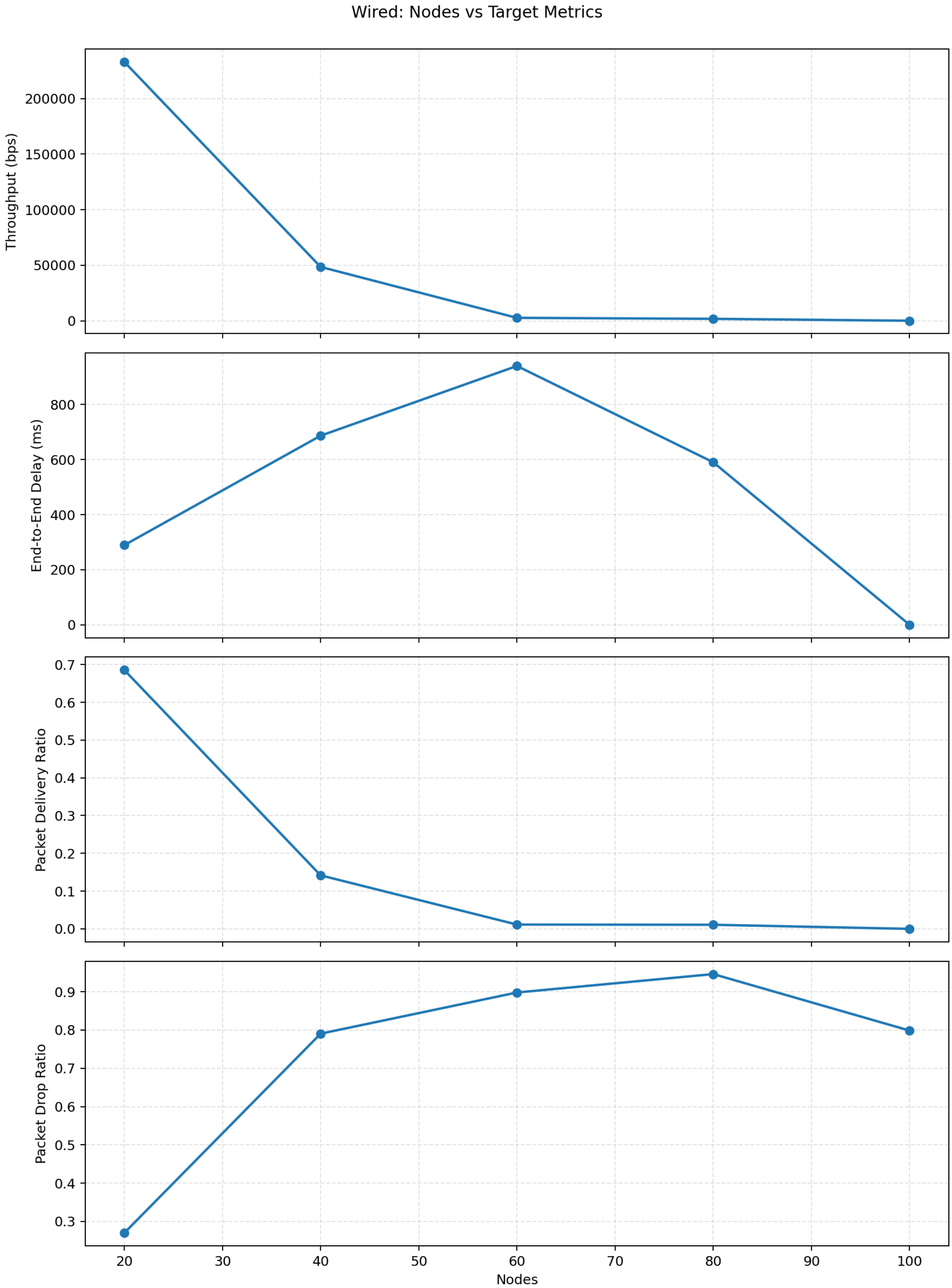

Figure 7: Wired: Performance vs Number of Nodes

Higher node counts result in severe degradation in throughput and reliability due to routing overhead and congestion.

6 Summary Findings

- AOMDV performs well under moderate load conditions.
- High PPS and flow counts lead to congestion and performance degradation.
- Mobility negatively impacts routing stability in wireless networks.
- Increasing nodes improves connectivity initially but degrades performance beyond a threshold.
- Wired networks show sharper performance drops under heavy load compared to wireless networks.

7 Discussion and Reflection

The results highlight the limitations of AOMDV under high traffic and dense network conditions. While multipath routing improves resilience, it introduces additional overhead that becomes significant under heavy load.

The deviation from the originally specified parameters was necessary due to:

- Hardware limitations preventing large-scale simulations
- Excessive congestion causing unstable or non-converging simulations

This adjustment ensured meaningful and stable results.
From the observations:

- There exists an optimal operating region for AOMDV
- Beyond this region, performance degrades rapidly
- Trade-offs between throughput, delay, and reliability are clearly visible

Future work could explore:

- Comparison with other routing protocols (AODV, DSR)
- Optimization techniques for congestion control
- Adaptive routing based on network conditions

8 Conclusion

This study demonstrates that AOMDV is effective in moderate network conditions but struggles under high load and dense deployments. Careful parameter tuning is essential to achieve optimal performance in both wired and wireless environments.